

Comprehensive Networking Assignment

IPv4 & IPv6 Addressing, Subnetting Calculations, and Verification

1. Key Differences Between IPv4 and IPv6 Addressing

IPv4 and IPv6 are the two principal versions of the Internet Protocol used to identify devices on a network. Below is a comparative overview of their core addressing characteristics:

Feature	IPv4 Addressing	IPv6 Addressing
Address Length	32-bit numeric address	128-bit alphanumeric address
Notation Format	Dotted-decimal (e.g., 192.168.1.1)	Colon-hexadecimal (e.g., 2001:db8::1)
Address Space	~4.3 billion unique addresses (2^{32})	~ 3.4×10^{38} unique addresses (2^{128})
Configuration	Relies on Manual configuration or DHCP/DHCPv4	Supports Stateless Address Autoconfiguration (SLAAC) & DHCPv6
Security (IPsec)	Optional add-on implementation	Built-in and mandatory across protocol suite
Traffic Types	Supports Unicast, Broadcast, and Multicast	Supports Unicast, Anycast, and Multicast (No Broadcast)

2. Home WiFi Static IPv4 Assignment

For a home network operating on the 192.168.10.0/24 network (Subnet Mask: 255.255.255.0, Default Gateway: 192.168.10.1), static IP addresses are assigned to four devices as follows:

Device	Assigned IP Address	Subnet Mask	Default Gateway
Laptop	192.168.10.10	255.255.255.0	192.168.10.1
Phone	192.168.10.11	255.255.255.0	192.168.10.1
Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
Printer	192.168.10.13	255.255.255.0	192.168.10.1

3. Subnetting Calculations for 192.168.1.0 /27

Given the base network 192.168.1.0 with a /27 CIDR mask (255.255.255.224):

- **Subnet Bits Borrowed:** $27 - 24 = 3$ bits
- **Valid Subnets:** $2^3 = 8$ subnets
- **Host Bits Remaining:** $32 - 27 = 5$ bits
- **Usable Hosts per Subnet:** $2^5 - 2 = 32 - 2 = 30$ hosts
- **Block Size (Magic Number):** $256 - 224 = 32$

First Three Valid Subnet Ranges:

Subnet	Network Address	Usable Host Range	Broadcast Address
Subnet 1	192.168.1.0	192.168.1.1 – 192.168.1.30	192.168.1.31
Subnet 2	192.168.1.32	192.168.1.33 – 192.168.1.62	192.168.1.63
Subnet 3	192.168.1.64	192.168.1.65 – 192.168.1.94	192.168.1.95

4. IP Address Analysis (10.0.5.17 / 255.255.255.0)

Given IP address 10.0.5.17 and subnet mask 255.255.255.0 (CIDR: /24):

- **Network Address:** 10.0.5.0 (calculated via bitwise AND between IP and Subnet Mask)
- **Broadcast Address:** 10.0.5.255
- **Valid Host Address Range:** 10.0.5.1 to 10.0.5.254

5. Online Subnet Calculator Verification

Verification Result: The manual calculations for Task 4 matched the online subnet calculator tool's output precisely, confirming a network address of 10.0.5.0, broadcast address of 10.0.5.255, and 254 usable host addresses.